# Hippocampal Replay Represents Sequential Relationship During a Temporal Sequence of Memory Task

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## Abstract

The **hippocampus** functions as the brain's central component for creating and organizing memories, demonstrated by its association with sequential spatial and temporal neural patterns. However, the neural mechanisms behind this phenomenon remain amid constant discussion. Our research investigates hippocampal ensemble activity during offline intervals of sequential events to search for evidence of **non-spatial replay**. We employ a **peri-stimulus time histogram** to visualize patterns in the neural spike data. We later extract the neural decoding probabilities for each odor using a **penalized multinomial regression model**. We then use **pointwise t-tests** to determine whether decoding probabilities for differing offline intervals have different patterns following out-of-sequence events.

## Background Info and Scientific Goals

- Five rats are trained to recognize a distinct sequence of five odors, then tasked to identify whether the odors presented are **in sequence** (InSeq; e.g. ABCDE) or **out of sequence** (OutSeq; e.g, ABDDE).
- Neural firing activity is recorded from the dorsal CA-1 region of the hippocampus.
- Rats receive a water reward for each correct evaluation and otherwise terminate the sequence with an incorrect evaluation.
- We aim to analyze CA-1 ensemble activity during offline intervals between odors for evidence of sequence replay.

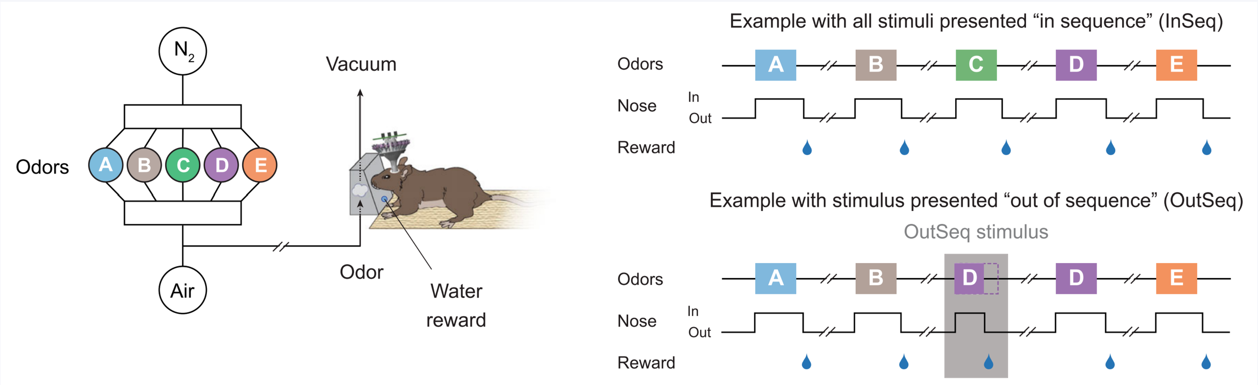


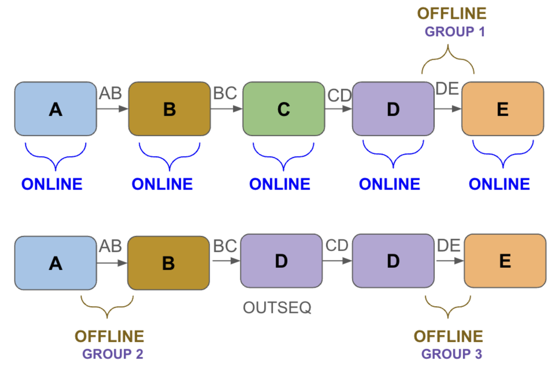

Figure 1. Rats are presented with odors released on the left. Odors are either in sequence or out of sequence, and rats must withdraw their nose within 1.2 seconds if an odor is out of sequence.

## Data Structure and Variables

The data provided by Fortin Lab includes two data sets:

- **Behavior Performance Data**: Time of odor release during sequences, rat performance, and information indicating InSeq or OutSeq.
- **Neural Ensemble Data:** Neural activity collected from spikes during odor release (online intervals) and periods between stimuli (offline intervals); used to test the models' accuracy and generalization.

**Offline intervals are split into three groups:**




## Exploratory Data Analysis

Before drawing conclusions from the rats' neural data, we first visualize how neurons respond to the presented odors. We also wrangle the data into a wide format to epoch trials into whole sequences, allowing for a comprehensive analysis of the neural response to odors presented.

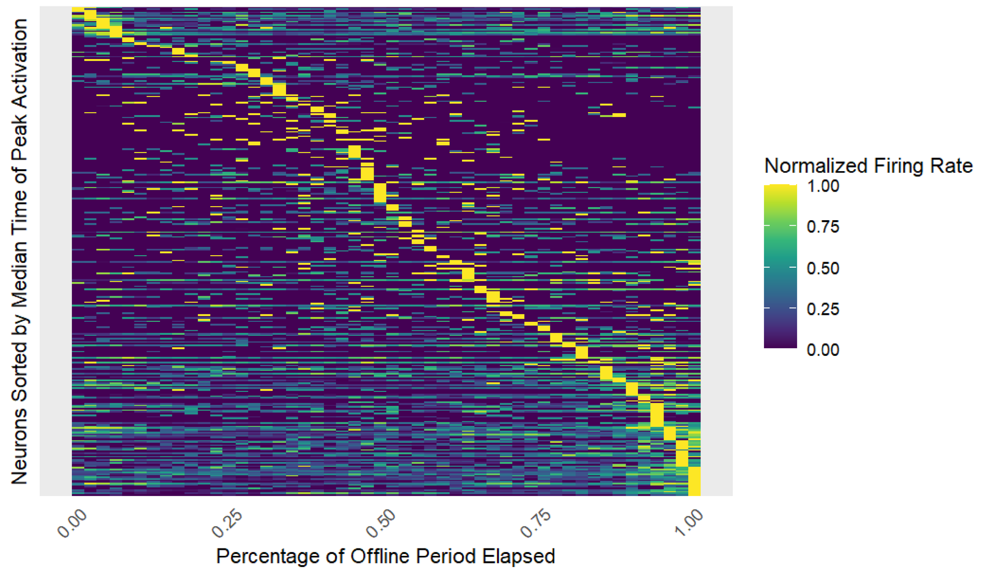



Figure 2. A peri-stimulus time histogram visualizing individual neuron firings during the A-B offline period for in-sequence sequences. Each row summarizes the firing rate of a single neuron for one rat for every 1/50th of the offline period.

## Modeling

Previous studies show that the CA-1 region of the hippocampus encodes spatial events in sequence. We investigate how our subjects recall non-spatial sequences, focusing on how in-sequence versus out-of-sequence trials affect neural activity and enhance understanding of both rat and human brains.

- We use an **elastic-net regularized generalized linear model (GLMNET)** for multinomial logistic regression with 10-fold cross-validation to decode ensemble activity.
- The model manages an elevated Type I error risk associated with analyzing high dimensional data from many neurons.
- It is trained using recorded online interval spike data, enabling it to link specific neuron firing patterns with particular odors.
- It then decodes neural firing patterns during offline periods and outputs the probability that each pattern is associated with each odor.

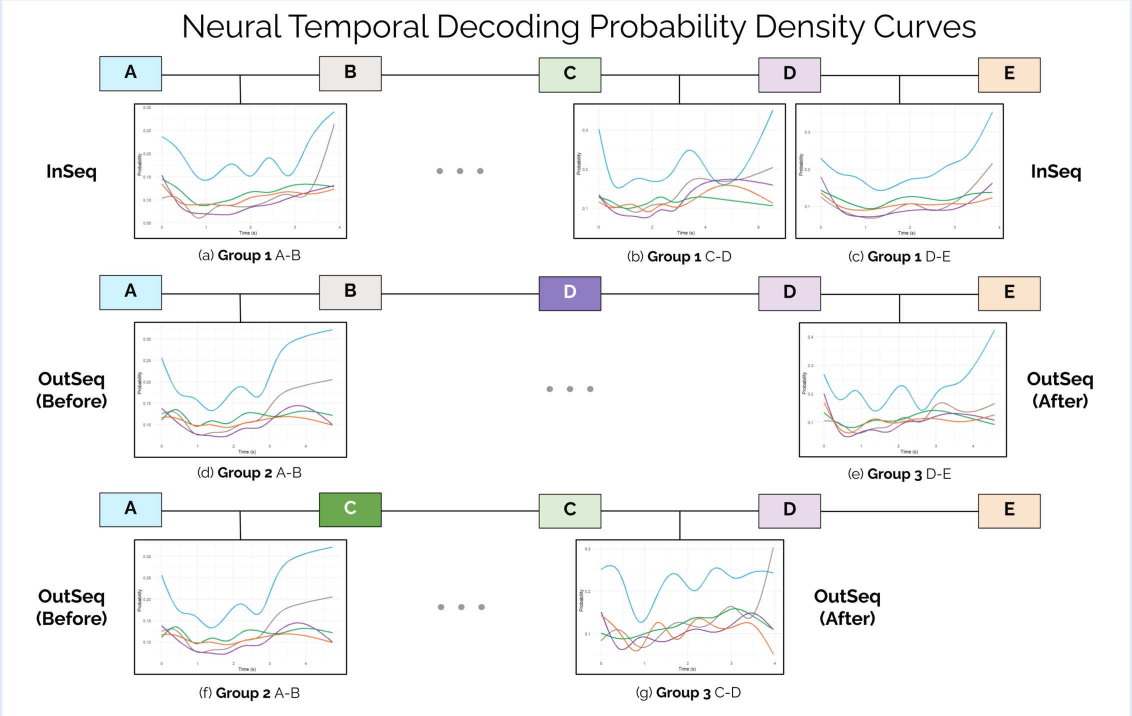



Figure 3. Median odor probabilities across sequences for one rat, separated by group and interval.

## Hypothesis Testing

We aggregate five odor probability curves into first principal components obtained from **principal component analysis (PCA)**. We apply functional data analysis with a **pointwise t-test** to analyze potential differences in neural activity **between groups 1 and 3**, also testing different intervals **within group 1** as a comparison.

**Our hypotheses are (with a significance level of 0.05):**

- $H_0$: Out-of-sequence odors ***do not*** significantly affect subsequent neural activity.
- $H_a$: Out-of-sequence odors ***do*** significantly affect subsequent neural activity.

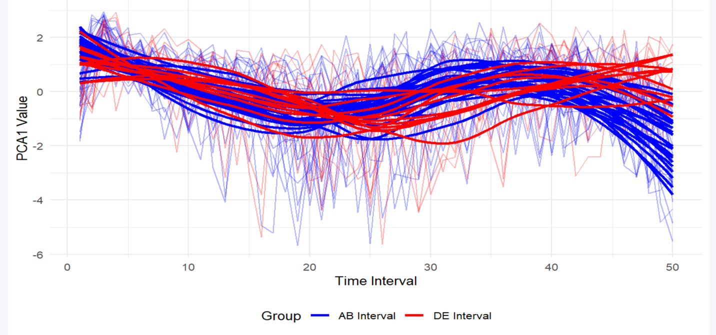


Figure 4. PCA distributions for group 1, intervals A-B and D-E.

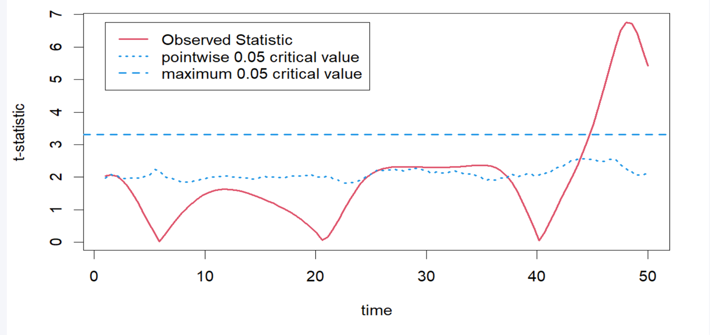



Figure 5. Observed t-stat and pointwise & maximum 0.05 critical values for group 1, intervals A-B and D-E.

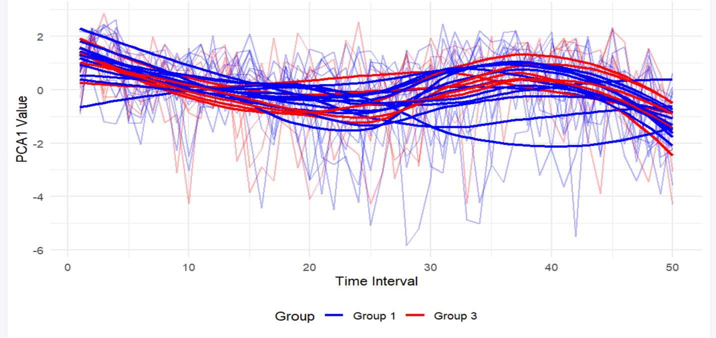


Figure 6. PCA distributions for group 1 and 3, interval C-D.

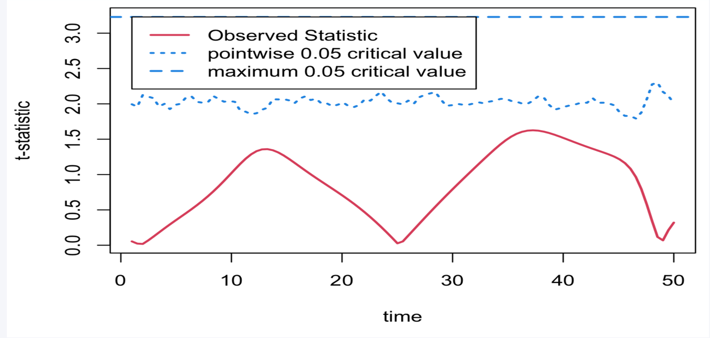



Figure 7. Observed t-stat and pointwise & maximum 0.05 critical values for group 1 and 3, interval C-D.

## Results and Future Work

**Within group 1**, our observed statistic **exceeds** our critical values at the end of the interval, meaning differing intervals within groups have different neural decoding patterns.

**Between groups 1 and 3**, our observed statistic **does not exceed** our critical values. We **do not have enough evidence to reject the null hypothesis** that out-of-sequence odors significantly affect subsequent neural activity.

**In the future**, we plan to sort neurons by their associated odors to expand the context of our PSTH. We also expect to extend our research to activity during online out-of-sequence events and the offline intervals that immediately follow.

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**References:** [1] Shahbaba, B., Li, L., Agostinelli, F. et al. Hippocampal ensembles represent sequential relationships among an extended sequence of nonspatial events. Nat Commun 13, 787 (2022). https://doi.org/10.1038/s41467-022-28057-6